

AI-POWERED EMPLOYEE OPERATIONS & PRODUCTIVITY AGENT

Consolidated Solution Proposal

A Specialised AI Agent for Email Automation and Workforce Productivity Insight — Not a Generic Chatbot

Prepared: August 2026

This document merges the original AI Email Agent Proof of Concept with the full Employee Operations & Productivity Agent proposal into a single, complete description of the platform, presented as one unified scope of work.

How to Read This Document

Every feature and integration below is tagged against one of four categories so it is clear exactly what is being committed to, and what depends on the client's own systems or approvals. Nothing described in this document has been built yet — this is a proposal for a build from scratch. The platform is presented as a single, unified scope of work rather than a series of separate delivery phases — items differ only in whether they are part of the core build, whether they depend on a client-side approval or access grant, or whether they are optional extensions.

Tag	Meaning
A — Core Build	To be built as part of the core scope, in the first phase of the build.
B — Client Dependency	Part of the committed scope, but requires access, approval or infrastructure from the client before work on it can start.
C — Optional / Future	Technically feasible, not committed as part of this scope, priced only if requested.
D — Out of Scope	Explicitly excluded, usually for privacy, legal or technical reasons.

1. What Is This Project?

In plain words: this is a specialised AI agent that reads a company's incoming Microsoft 365 (Outlook) emails, figures out what each email is about, checks who sent it, decides — using rules the business owner sets, not the AI — what should happen next, assigns it to the right person or team with a deadline, and, when it is safe to do so, sends an automatic reply. Every decision it makes is written down so it can be checked later.

That original requirement has grown into a broader need: a single, specialised agent that automates email handling and also gives management a privacy-conscious, evidence-based view of how work actually happens across email, Teams, WhatsApp, browsers, desktop applications and the client's ERP. This document keeps the email-agent architecture intact and extends it with an Employee Activity Agent, a Productivity Analytics layer, and a controlled Natural-Language Management Agent — delivered as one product, not a bundle of unrelated integrations.

This is a proof-of-concept-to-production proposal, not a finished production system. Several capabilities that were discussed — full Teams message content, personal WhatsApp monitoring, guaranteed fully local AI — are either technically unrealistic, privacy-inappropriate, or dependent on approvals not yet granted. These are called out explicitly rather than implied as included.

2. Problem Statement

Email volume and cross-channel communication (Teams, WhatsApp, browser-based tools) are growing faster than the business's ability to manually triage, respond to and report on them. There is no single, reliable view of response times, workload, or how time is actually spent during the working day, which makes capacity planning and fair performance evaluation difficult.

3. Current Business Challenges

- Emails are read, triaged and assigned manually; this does not scale and produces inconsistent response times.
- No central view of incoming, pending and responded emails across the business.
- Important emails (VIP clients, RFPs, complaints) are not consistently prioritised.
- No structured, cross-application visibility into how employee time is spent.
- Monthly productivity or activity reporting is compiled manually today.
- RFP-related email is not systematically tracked from receipt through to response.
- Off-the-shelf suites (Microsoft 365, Google Workspace, Teams) provide messaging and isolated activity logs individually, but nothing that unifies communication automation with cross-application productivity analytics under organisation-specific rules.
- Any monitoring solution must handle sensitive communication and activity data carefully, and must be defensible from a privacy and compliance standpoint.

4. Proposed Solution

A specialised AI agent — branded and positioned as the client's own operations agent, not a general-purpose assistant — built around one architectural principle carried over from the original Email Agent design and applied consistently across every module:

AI UNDERSTANDS → POLICY ENGINE DECIDES → CODE EXECUTES → AUDIT LOG RECORDS

The AI is used to read, summarise, classify and draft — it never silently decides a business outcome. Routing, monitoring permissions, automation eligibility and reporting logic remain governed by deterministic policy code that management configures and controls. This split matters because it means the system behaves predictably and safely even though it uses AI: the AI cannot accidentally promise something, assign the wrong SLA, or reply to someone it shouldn't — those decisions are locked inside normal, testable code.

5. How It Works — In Simple Terms

Think of it as an assembly line. Every single email that arrives goes through the same fixed sequence of stations, one after another. Nothing is skipped and nothing happens out of order.

1. Email arrives in Microsoft 365 (Outlook)
2. System checks: “Have I already seen this exact email?”
3. System checks: “Do I know this sender?”
4. System reads earlier messages in the same conversation
5. AI reads the email and explains what it is about
6. Safety check: any legal threat or fraud claim?
7. Company's own rules (policies) decide what happens
8. Email is routed to the right team or person, with a deadline
9. A task is created for that person/team

- 10.If allowed, an automatic reply is sent
- 11.Everything that happened is logged for audit

6. Technology Stack

The platform is built using widely-used, production-grade web technologies. Nothing exotic — every piece was chosen because it is reliable and well documented.

Layer	Technology	Why It Is Used
Frontend	React	Builds the dashboard, task list, policy editor, contacts, activity timeline and audit-log screens as an interactive website.
Backend	FastAPI (Python)	Handles all requests from the frontend, runs the email and activity pipelines, and talks to the database.
Database	PostgreSQL (hosted on Neon)	Stores users, connected accounts, contacts, policies, teams, emails, activity events, tasks and audit logs.
Database migrations	Alembic	Safely updates the database structure over time without losing data.
Authentication	JWT (JSON Web Tokens)	Confirms who is logged in and keeps each user's data separated from every other user.
Email access	Microsoft Graph API + OAuth 2.0	Lets the system read and send email on the user's behalf without ever seeing their Microsoft 365 password.
AI / Language Model	OpenAI (GPT-4-class model)	Reads each email and activity signal and returns a structured understanding — category, urgency, intent and safety flags.
Background automation	APScheduler	Runs the email-checking and activity-capture pipelines automatically every few minutes for every connected source.
Token/credential security	Encryption at rest	Access and refresh tokens are stored encrypted in the database, never in plain text.

6.1 FastAPI (the Backend)

FastAPI is the control centre. Every action in the app — logging in, connecting a source, saving a policy, viewing the dashboard — is a request that the React frontend sends to FastAPI. FastAPI checks who is asking (via the JWT token), fetches or updates the right data in PostgreSQL, and sends a response back. It also hosts the background job that processes emails and activity events automatically.

6.2 React (the Frontend)

React is the website itself — the part the user clicks on. It shows the dashboard with numbers (how many sales emails today, how many are overdue, how time is being spent), lets the user create contacts, teams and policies, and lists tasks that need attention. It never talks to Microsoft 365, other sources, or the AI directly; it only talks to FastAPI, which does the real work.

6.3 PostgreSQL (the Database)

This is where everything is permanently stored: which users exist, which accounts belong to which user, the list of contacts and their importance level, the policies that define business rules, the teams and their members, every email and activity event processed, every task created, and a full audit trail. Alembic is the tool used to change the database's structure safely as the project grows, without deleting existing data.

6.4 Microsoft Graph API + OAuth 2.0

Instead of asking a user for their Microsoft 365 password (which would be unsafe), the system uses OAuth — the same “Sign in with Microsoft” style flow used by most trusted apps, via Azure AD / Microsoft Entra ID. The user logs into Microsoft 365 directly and grants permission. Microsoft then gives the system a secure access token and a refresh token, which are encrypted and stored. These tokens are what let the system read new emails and send replies on the user's behalf, and they can be revoked by the user at any time from their Microsoft 365 account (Entra ID). The same OAuth pattern is used for every other connected source (Teams, Calendar) once its access is granted.

6.5 OpenAI (the AI)

OpenAI is the service that runs the language model used by the platform. For every new email, the system sends the email's text (plus a short summary of earlier messages in the same thread) to the model, along with the list of categories the business owner has configured. The model replies with a structured result: the category, how urgent it seems, whether it looks like a legal threat or fraud claim, an RFP flag where relevant, and how confident it is. The AI is deliberately never asked to decide what action to take — only to describe what the email is. The same principle is applied to activity narratives: the AI explains trends the deterministic engine has already calculated, it never invents a score.

6.6 APScheduler (Background Automation)

This is a scheduler that runs inside the backend and wakes up on a fixed interval (for example every 1–2 minutes). Each time it wakes up, it goes through every connected account and every connected activity source, one at a time, and runs the full pipeline for any new messages or events. This is what makes the system “automatic” — no person has to click a button.

6.7 JWT Authentication & Multi-User Isolation

When a user logs in, they receive a signed token (JWT) that proves who they are on every future request. The backend uses this token to make sure a user can only ever see their own connected accounts, contacts, policies, teams, tasks, activity data and audit logs — never another user's. This is what allows the same system to serve many separate companies or teams at once, safely.

7. Product Vision

“AI-Powered Employee Operations & Productivity Agent” — one platform, ten modules, a single normalised view of communication and activity data, with the Email Agent as its first module to be built.

#	Module	Purpose
1	AI Email Agent	Reads, classifies, summarises and drafts/sends email under policy control (the foundational module, built first).

#	Module	Purpose
2	Employee Management	Add/remove employees, assign policies, control per-source monitoring consent.
3	Policy & Workflow Engine	Deterministic rules for routing, SLA, automation eligibility and monitoring scope.
4	Employee Activity Agent	Usage/time-spent capture from email, Teams, WhatsApp Business, browser and desktop.
5	Productivity Analytics	Deterministic metrics first, AI-generated narrative insight second.
6	Management Dashboard	Unified view across all modules for admins and managers.
7	AI Management Agent	Controlled natural-language querying and action-taking, tool-gated and audited.
8	Integration Hub	Common ingestion/normalisation layer for every connected source.
9	Security, Privacy & Audit	RBAC, consent, encryption, audit trail, data retention.
10	Voice Interface	Text-to-speech briefing and speech-to-text instruction entry.

8. Solution Architecture

Every data source is treated as an input to one common pipeline rather than as a standalone integration. This is what allows email activity, meeting activity and application usage to appear side-by-side on the same report. Each source adapter converts its native data (an Outlook email message, a Teams meeting record, a browser tab event) into the same activity/event shape before it reaches analytics or the AI layer. This is what makes the platform “one agent with many senses” rather than several disconnected integrations glued together.

Sources (Microsoft 365/Outlook, Teams, WhatsApp Business, Calendar, Browser, Desktop/OS, ERP) feed into an Integration Layer (per-source adapters, OAuth/API, token management, health checks), which writes into a Normalised Activity/Event Model (employee_id, source, activity_type, start_time, end_time, duration, metadata). This model feeds the Activity & Analytics Engine (deterministic metrics: response time, SLA, usage, RFP), which in turn feeds two consumers side by side: the AI Agent Layer (summarise / classify / draft / explain trends / tool-gated natural-language query) and the Management Dashboard (screens, reports, integration health, policy management). Every action taken by either consumer passes through the Policy Engine, then Action Execution, then the Audit Log.

9. Core Modules

Expanded description of each module, with its primary responsibility and its main dependency.

Module	Responsibility	Primary Dependency
AI Email Agent	Ingest, classify, summarise, draft/send email; SLA and routing.	Microsoft 365 OAuth (Email Agent module).
Employee Management	Add/remove employees; per-source consent and policy assignment.	Admin role in platform.
Policy & Workflow Engine	Routing rules, automation eligibility, monitoring	None — core, in-house.

Module	Responsibility	Primary Dependency
	scope.	
Employee Activity Agent	Capture usage/time events from connected sources.	Per-source API access (client-dependent).
Productivity Analytics	Deterministic KPIs plus optional AI narrative.	Activity Agent data.
Management Dashboard	Present all of the above to admins/managers.	None — core, in-house.
AI Management Agent	Controlled natural-language query/action interface.	Policy Engine, tool registry.
Integration Hub	Adapters, normalisation, token/health management.	Per-source credentials.
Security, Privacy & Audit	RBAC, encryption, consent, audit, retention.	Client data-handling policy.
Voice Interface	TTS playback, STT instruction capture.	Browser/device mic & speaker access.

10. Detailed Features — Classification Matrix

This is the coverage checklist for the full requirement, for the platform to be built. Every requested capability appears exactly once below, tagged per the legend at the start of this document, so nothing is silently included or silently dropped.

Feature	Class	Notes
Central email dashboard (incoming/pending/responded)	A	Built as the core dashboard module.
AI email summarisation	A	Built as a core capability.
AI-drafted replies (manual review)	A	Built as a core capability.
Policy-gated automatic replies	A	Safety-gate logic built in from the start.
Configurable automated vs manual buckets	A	Built as part of the policy engine.
Multiple policies assignable to specific users/teams	A	Built as part of the policy engine.
Secure per-user email account connection	A	Built as per-user OAuth.
Add/remove employees from monitoring	A	New Employee Management module.
RBAC (Super Admin/Admin/Manager/Employee)	A	New — see role table.
Email activity metrics (received/responded)	A	Deterministic, calculated from email data.
RFP detection and tracking	A	Built as an extension of the AI classification, with an RFP flag.
Monthly productivity report (email + connected sources)	A	Scope grows as each source is connected.
Deterministic productivity KPIs	A	Calculated by code, not the AI.

Feature	Class	Notes
Integration health monitoring	A	Live for every connected source.
Audit log of all actions	A	Built once and extended to every new module.
Browser usage/time tracking (extension)	B	Requires a lightweight browser extension.
Desktop/OS application usage tracking	B	Requires a lightweight desktop agent.
Microsoft Teams activity metadata (meetings/calls/duration)	B	Depends on Graph API permissions.
Calendar integration	B	Depends on calendar API access.
WhatsApp Business activity/messaging	B	Depends on WhatsApp Business API approval.
Client ERP integration	B	Depends on ERP API documentation/access.
Text-to-speech for important emails	B	Voice module.
Speech-to-text for instructions	B	Voice module.
Natural-language management agent (read-only queries)	B	Depends on Productivity Analytics being live.
Natural-language management agent (controlled actions)	B	Ships after read-only queries are validated; confirmation-gated.
AI-generated trend narratives on top of KPIs	C	Optional enhancement once KPI data exists.
Cross-department configurable productivity definitions	C	Optional, needs client input per department.
Local/on-prem LLM deployment	C	Evaluated for production.
Idle/active time where OS-supported	C	Best-effort, OS-dependent, not guaranteed cross-platform.
Continuous screen recording	D	Explicitly excluded — not part of this solution.
Keystroke logging	D	Explicitly excluded.
Personal WhatsApp message monitoring	D	Explicitly excluded — only WhatsApp Business API activity is in scope.
Full Teams message content monitoring	D	Explicitly excluded — metadata only.
Custom ERP-side development	D	Excluded from this proposal; would be scoped separately.

11. Integration Architecture

Each source is captured only at the level described below — usage and metadata, not content — consistent with the client's own stated preference for usage/time analytics rather than surveillance.

Source	Captured	Explicitly NOT Captured	Class
Microsoft 365	Message metadata, thread context, AI-	N/A — per-user OAuth, once	A

Source	Captured	Explicitly NOT Captured	Class
(Outlook)	derived category/urgency; send/reply actions under policy.	consented.	
Microsoft Teams	Meeting/call occurrence, duration, timestamps, participation metadata.	Chat/message content, call audio, screen shares.	B
WhatsApp Business	Business-account message counts, response times, activity timestamps (Business API only).	Any personal WhatsApp account, message content beyond what the Business API exposes.	B
Calendar	Event timestamps, duration, meeting counts.	Meeting content/notes.	B
Browser	Domain/site visited, duration, timestamp, device identifier.	Page content, form data, keystrokes, screenshots.	B
Desktop/OS	Active application name, window focus duration, start/end time.	Screen content, keystrokes, file contents.	B
Client ERP	Defined by client API — e.g. RFP or order-related events relevant to reporting.	Anything outside the agreed API scope.	B

Supported browsers and operating systems for the lightweight agents will be confirmed against the client's actual device fleet before that work is scoped in detail — universal OS/browser coverage is not committed in this document.

12. Email AI Agent — Detailed Workflow

This module is the original AI Email Agent, carried forward unchanged in principle and extended with RFP tracking and richer policy assignment.

12.1 Setup (done once by the business owner)

1. Register and log in — the system issues a JWT so the backend always knows which user it is talking to.
2. Connect Microsoft 365 (Outlook) using OAuth — no password is ever shared with the app.
3. Add contacts — e.g. a VIP customer with a 2-hour SLA, or a normal customer with a 24-hour SLA.
4. Create teams — e.g. Sales Team, Billing Team, Support Team, Management/Human Review.
5. Define email categories — e.g. Sales, Billing, Complaint, Legal, General Inquiry. These become the fixed vocabulary the AI is allowed to choose from.
6. Create policies — plain “if this, then that” business rules, e.g. “If contact is VVIP → priority HIGH, SLA 2 hours, route to Management, no auto-reply.”

12.2 Live Processing (runs automatically for every incoming email)

Ingest: a scheduler polls each connected Microsoft 365 mailbox for new messages via Microsoft Graph; duplicate detection via the unique message ID ensures the same email is never processed twice; sender, subject, body, thread ID and date are extracted and cleaned.

Context: the sender is looked up against known contacts (pulling in classification such as VVIP and a default SLA where known); earlier messages in the same thread are retrieved so the AI understands what is being followed up on, and the system checks whether a task already exists for this thread.

Understand: the email text, thread context and the business's own category list are sent to the AI, which returns category, urgency, confidence, legal/fraud flags, and an RFP flag — it does not decide the action.

Safety gate: legal threats, fraud claims or low confidence force human review, overriding any policy and cannot be bypassed.

Decide: the policy engine matches the email against configured rules in priority order and stops at the first match; routing and SLA are calculated deterministically from that match or from the contact's own assigned rule.

Act: a task is created for the assigned team/person, or an existing task on the same thread is updated instead of duplicated; a reply is sent only if policy, contact status, confidence and safety checks all allow it, drafted by the AI and sent from the user's own Microsoft 365 account via the Microsoft Graph API.

Audit: every step above is logged with a timestamp, so the entire decision path can be reviewed later, and the dashboard refreshes to reflect the new activity.

12.3 Advanced Communication Orchestration Workflow

Include:

- Sender Validation
- Contact Classification
- VVIP / VIP / Class A-B-C priority engine
- Unknown Sender Handling
- Historical Email Thread Analysis
- SLA Assignment
- Task Creation
- Regional Routing
- Auto-reply Eligibility
- Escalation Management
- Dashboard Visibility

Key business flow:

Email Arrives->Validate Sender->Known / Unknown Contact Check ->Priority Check ->Intent Classification ->Historical Conversation Review ->Policy Engine Match -> SLA Assignment ->Regional/Functional Routing ->Task Creation -> Escalation Rules -> AI Draft Reply->Human Review or Auto Send -> Audit Trail -> Dashboard Update

12.4 Contact Priority Framework

Priority	Example SLA
Board Member	Immediate
Executive Management	1 Hour
VVIP Customer	2 Hours
VIP Customer	4 Hours
Class A Customer	4 Hours
Class B Customer	8 Hours
Class C Customer	24 Hours
Supplier	24 Hours
Unknown Contact	Human Review

13. Worked Example

To make the workflow concrete, here is one real email moving through the whole system.

Incoming email — From: a known contact's business address · Subject: "Urgent quotation request" · Body: "We need a quotation for 500 units, please send ASAP."

Step	What Happens
Contact lookup	The sender is found in Contacts and is classified as VVIP.
Thread check	This is part of an existing conversation, so earlier messages are pulled in.
AI classification	Category = Sales, Urgency = HIGH, Confidence = 0.96, Legal threat = No, Fraud = No.
Safety gate	No issues found — processing continues normally.
Policy match	The VVIP policy matches: Priority = HIGH, SLA = 2 hours, Route = Management, Auto-reply = No.
Routing	The email is assigned to the Management team.
Task created	Task #125 — Priority HIGH, due in 2 hours, status OPEN, assigned to Manager.
Audit log	Fetch, contact lookup, classification, policy match, routing and task creation are all recorded.
Dashboard	Management sees one new VVIP Sales email, due in 2 hours.

A second, contrasting example: an email from an unknown sender making a fraud allegation is classified by the AI as a "Complaint", but the safety gate detects the fraud/legal flags, overrides any normal policy, blocks any auto-reply, and routes the email straight to Human Review with a task created for a person to handle it manually.

14. What the AI Does vs. What Plain Code Does

This separation is the safety backbone of the whole system, so it is worth stating clearly. The AI explains, and the code decides — the AI can never independently promise a refund, hand a task to the wrong department, or skip a legal review, because those actions are hard-coded business rules the business owner controls directly through the Policies screen.

Handled by the AI	Handled by Deterministic (Non-AI) Code
Understanding what the email is about	Looking up the sender in the contact list
Classifying it into one of the business's categories	Matching the business's own policies
Judging urgency and intent	Deciding routing (which team/person)
Detecting legal/fraud/regulatory language	Calculating the SLA due date
Drafting a reply when one is allowed	Creating and de-duplicating tasks
Explaining productivity trends from calculated numbers	Enforcing safety overrides and human review
	Keeping each user's data isolated and secure
	Writing the audit log

15. Employee Activity Agent

Captures usage/time-spent events only, per source, as defined in the Integration Architecture. Monitoring is opt-in per employee and per source, controlled entirely by Admin/Manager roles through Employee Management.

- No continuous screen recording, at any stage.
- No keystroke logging.
- No capture of message or document content beyond what is explicitly listed as “captured” for each source.
- Monitoring can be switched on/off globally per employee, and independently per source (e.g. email tracked, browser not).
- Legal and regulatory requirements for workplace monitoring vary by jurisdiction and company policy; the client is responsible for confirming monitoring is compliant with applicable law and internal policy before this capability goes live.

16. Unified Activity / Event Model

Every source adapter emits the same event shape, which is what allows one analytics engine and one dashboard to serve every source without source-specific logic downstream.

Field	Description
employee_id	Internal identifier linking the event to a specific employee record.
source	Origin system, e.g. gmail, teams, whatsapp_business, browser, desktop, erp.
activity_type	Normalised type, e.g. email_received, email_responded, meeting, app_usage,

Field	Description
	site_visit.
start_time / timestamp	When the activity began or occurred.
end_time	When the activity ended (where applicable).
duration	Derived or reported length of the activity.
metadata	Source-specific detail limited to what the Integration Architecture defines as captured (e.g. domain, app name, RFP flag).

Example (illustrative, not a literal storage format): employee_id "E-104", source "browser", activity_type "site_visit", start_time 09:14:02, end_time 09:26:40, duration_sec 758, metadata { domain: erp.client-portal.com }.

17. Productivity Analytics

Activity is not the same as productivity, and the platform does not let the AI decide productivity scores. Deterministic metrics are calculated first; AI is used only afterwards, optionally, to explain trends in plain language for management.

17.1 Deterministic metrics (calculated by code, not the AI)

- Emails received / responded, average response time, SLA compliance, pending backlog.
- Meeting duration and count, from Teams/Calendar.
- Application and site usage time by category, from desktop/browser agents.
- Active vs idle time where the OS reliably exposes it (best-effort).
- RFP emails received, responded, pending, and average RFP response time.

17.2 AI layer (optional, on top of the metrics above)

Plain-language narrative of trends, e.g. "response times rose 18% this month, concentrated in the Billing category" — phrased from the underlying numbers, never invented.

17.3 Configurability

What counts as "productive" application/site usage is configurable per department, since a browser-based CRM tab is productive for Sales and neutral for Engineering. This mapping is defined by the client, not assumed by the platform.

18. Policy & Workflow Engine

The same engine that routes email is also used to govern monitoring scope and automation eligibility for the newer modules, so there is one place — not several — where business rules live.

Policy Type	Example
Email routing/SLA	"If contact is VVIP → priority HIGH, SLA 2 hours, route to Management, no auto-reply."

Policy Type	Example
Automation eligibility	"Auto-reply allowed for Category=General Inquiry only if confidence ≥ 0.9 ."
Monitoring scope	"Employee E-104: email + browser monitored; Teams and desktop not enabled."
Escalation	"If RFP pending response > 48 hours $\rightarrow$ notify Sales Manager."

19. Natural Language Management Agent

A controlled interface for management to query data and trigger permitted actions in plain language — never a free-form code or SQL execution channel. The pipeline is: natural language $\rightarrow$ intent detection $\rightarrow$ permission check $\rightarrow$ tool selection $\rightarrow$ policy validation $\rightarrow$ confirmation (if sensitive/irreversible) $\rightarrow$ action execution via existing controlled tools only $\rightarrow$ audit log.

- "Show employees whose average email response time exceeded 4 hours." — read-only.
- "Show pending high-priority emails." — read-only.
- "Assign Rahul the Sales Priority Policy." — action, requires confirmation.
- "Draft a response to the latest RFP." — action, routed through the Email Agent's draft flow.

The AI never generates or executes arbitrary system commands or SQL. Every action is routed through a fixed, pre-approved set of tools, each of which independently enforces the same policy and permission checks as the rest of the platform. Sensitive or irreversible actions (e.g. removing an employee from monitoring, sending an email) require explicit confirmation.

20. Voice Interface

Text-to-speech for reading important emails aloud, and speech-to-text for issuing instructions to the agent — a convenience layer on top of the Email Agent and Management Agent, not a separate product.

- Browser-based TTS/STT using standard web APIs; no dedicated hardware assumed.
- Accuracy and language support depend on the underlying speech service selected and should be validated against the client's accents/languages before rollout.

21. Data Model (What Is Stored)

Each registered user owns their own complete, isolated set of the following records:

- Connected accounts — inbox and other source connections, encrypted access/refresh tokens, token expiry.
- Contacts — sender email, classification (e.g. VVIP, CLASS_A, CLASS_C), default SLA, assigned person.
- Email categories — the fixed vocabulary the AI is allowed to classify emails into.
- Policies — the "if this, then that" business rules, checked in priority order.
- Teams and team members — the groups that emails can be routed to.
- Emails — every processed message, with its Microsoft Graph message ID (to prevent duplicates), thread ID, and AI classification.
- Activity events — the normalised event records described in the Unified Activity/Event Model.

- Tasks — one per email or email thread, with priority, SLA, due date, assignee and status.
- Audit logs — a permanent, timestamped record of every decision made.

Because every one of these tables is linked back to a specific user, one company's emails, contacts, policies, activity data and tasks can never be seen by another company using the same platform.

22. Security & Privacy

- JWT authentication for every request.
- OAuth for every third-party connection — the platform never sees or stores a user's password.
- Credentials and tokens encrypted at rest, across all integrations.
- Full tenant/user data isolation, covered by the automated test suite and extended to every new table.
- Data minimisation: only the fields defined in the Integration Architecture and the Unified Event Model are stored; no incidental content capture.
- Configurable data retention period for activity events and reports.
- Global and per-source monitoring on/off switches, controlled by Admin/Manager roles only.
- Legal threats and fraud allegations always force human review, regardless of any auto-reply policy configured — this cannot be switched off by a normal policy.
- Unknown senders (not in the Contacts list) can never receive an automatic reply.

22.1 Roles

Role	Can Do	Cannot Do
Super Admin	Everything below, plus manage other admins and org-wide settings.	—
Admin	Add/remove employees, create/assign policies, connect integrations, view all analytics and audit logs, export reports.	Manage other admins' accounts.
Manager	View their team's analytics and reports, assign existing policies to their team, view relevant audit entries.	Create global policies, connect new integrations, view other teams' data.
Employee	View their own activity summary (transparency), connect their own email/Teams/etc. when asked.	View others' data, change policies, disable monitoring on themselves.

23. Integration Health

Each connection is monitored so admins know immediately when a source stops reporting data, rather than discovering a silent gap in a monthly report.

Source	Status (example)	Detail Tracked
Microsoft 365 (Outlook)	✓ Connected	Last successful sync time, token expiry countdown.
Microsoft Teams	✓ Connected	Last sync, Graph API permission status.

Source	Status (example)	Detail Tracked
Calendar	✓ Connected	Last sync.
Browser extension	⚠ Disconnected	Last seen, device/employee affected.
Client ERP	✗ Error	Last error message, retry count, admin alert sent.

- Automatic retry with backoff for transient failures; admin notification after repeated failures.
- Manual reconnection flow for expired OAuth tokens, surfaced directly on this screen.

24. Dashboard & UX

Wireframe-level structure for each screen; visual design is produced and reviewed with the client before development, but the content of each screen is defined now so scope is unambiguous.

24.1 Management Overview

Emails today, open tasks, overdue, auto-replies sent, average response time, SLA compliance %, active employees, and an integration health strip (Microsoft 365 OK, Teams OK, ERP ERROR).

24.2 Email Command Centre

Incoming/Pending/Responded tabs with filters for category, urgency and RFP; a selected-email panel showing the AI summary, a suggested reply with Edit/Approve & Send, and voice controls (Play summary, Speak reply).

24.3 Employee Management

Employee list with Add Employee; each row shows name, policies assigned, sources monitored and status, with row actions to assign a policy, toggle sources, or remove.

24.4 Policy Management

Policy list with New Policy; a rule builder (IF condition THEN routing/SLA/automation); assignment to an employee/team multi-select.

24.5 Employee Activity Timeline

Employee and date-range selectors, with a timeline view (e.g. 09:00 Email, 09:15 Browser, 10:00 Meeting).

24.6 Productivity Analytics

KPI cards for response time, SLA %, usage by category and RFP; a weekly/monthly trend chart; an optional AI narrative panel.

24.7 Reports

Monthly productivity report generation and export (PDF/CSV), with report history and scheduled delivery settings.

24.8 Integration Health

A table of source, status, last sync and available reconnect action.

24.9 AI Management Agent

A chat-style input for natural-language questions, with results shown as a table plus a Confirm Action control where required.

24.10 Executive Communication Control Tower

Includes:

- VIP Communication Queue
- SLA Breach Monitor
- Escalated Emails
- Customer Response Obligations
- Regional Workloads
- Team Performance
- Pending Approvals
- High-Risk Communications
- RFP Pipeline
- Communication Backlog

25. Key User Journeys

25.1 Setup & Administration

1. Admin adds an employee in Employee Management.
2. Employee connects their Microsoft 365 account via OAuth.
3. Employee connects Teams/Calendar via OAuth once tenant permission is granted.
4. Admin assigns one or more policies to the employee.
5. Admin enables/disables monitoring globally and per source for that employee.

25.2 Email Processing

1. An incoming email is fetched and parsed.
2. The AI generates a summary and classification, including an RFP flag where relevant.
3. The policy engine determines routing, SLA and automation eligibility.
4. A human approves the draft where policy requires manual review; otherwise the reply is sent automatically.

25.3 Activity & Reporting

1. Connected sources emit activity events into the normalised event model.
2. The dashboard aggregates events into the employee timeline and KPI cards.
3. A monthly report is generated automatically and made available for export.

25.4 Management Agent

1. A manager asks a natural-language analytics question and receives a read-only answer.
2. A manager performs a controlled action (e.g. policy assignment) through the agent, with confirmation before it executes.

26. Why This Solution Is Different From Existing Market Products

Google Workspace with Gemini, Microsoft 365 with Copilot and Viva, and dedicated productivity-monitoring platforms such as ActivTrak or Hubstaff already provide substantial AI, analytics and automation individually. This platform is not claiming those products lack AI, cannot automate email, or cannot report on activity — they can, within their own ecosystems. The differentiation is specialisation, unification, customisation and control: one organisation-specific agent operating across all of those systems at once, governed by the client's own rules.

26.1 Cross-platform management intelligence

Productivity suites are strongest within their own ecosystem. This platform is designed as an intelligence layer across Microsoft 365 (Outlook), Microsoft Teams, Calendar, WhatsApp Business, browser, desktop applications, the client's ERP and other approved systems — all normalised into the common event model. The goal is not to replace Microsoft 365, Outlook or Teams; it is to sit across them.

26.2 A specialised agent, not a general-purpose assistant

The agent is purpose-built around this client's employees, departments, roles, policies, SLAs, email categories, RFP workflows, approval rules, automation permissions, productivity definitions and ERP workflows — it operates inside the client's own operating model rather than behaving as a generic conversational assistant.

26.3 Client-specific policy engine

The core control loop — AI understands, the deterministic policy engine decides, controlled code executes, the audit log records — lets management change business rules (e.g. “RFP from a VVIP customer → Sales Management, 2-hour SLA”, or “legal-risk language → no auto-reply, human review”) without retraining any model.

26.4 Email automation and workforce analytics in one platform

Google/Microsoft suites are strong at communication and collaboration with AI built in; dedicated monitoring platforms such as ActivTrak are strong at application, website and time analytics. This platform combines email intelligence, email automation, policy management, employee activity analytics, productivity reporting, cross-platform integrations and a management AI agent in one operational view, rather than requiring management to move between two categories of product.

26.5 Business-process-aware AI

The AI does more than summarise: an incoming RFP is identified, matched to the customer, prioritised, checked against policy, assigned to an employee or team, given an SLA and task, drafted for response, routed for human approval where required, logged, and folded into management analytics — one connected workflow rather than an isolated AI reply.

26.6 Privacy-conscious workforce analytics

The design favours application usage, website/domain usage, meeting duration, email activity, communication activity and active/idle time where technically available, and explicitly excludes continuous screen recording, keystroke logging, camera monitoring and unnecessary capture of private content. This is a deliberate architectural and privacy choice for this platform, focused on how work is distributed rather than continuously recording what an employee is doing.

26.7 A configurable productivity model, not a first-of-its-kind one

Productivity analytics already exist in the market; the differentiation is a business-specific configurable model — e.g. CRM/ERP/customer communication counted as productive for Sales, IDE/Git/Jira for Development, HRMS/email/meetings for HR — defined by the client, not assumed by the platform. Activity is not productivity: objective activity metrics are calculated first, configurable business rules are applied second, and AI explains the resulting insight rather than inventing a score.

26.8 Controlled natural-language actions

Management can ask questions and trigger permitted actions in plain language (“Show employees whose average email response time exceeded four hours this month”, “Assign Rahul to the Sales Priority Policy”) through the intent-detection → permission-check → tool-selection → policy-validation → confirmation → action → audit pipeline. This is controlled enterprise agent functionality, not unrestricted AI automation.

26.9 AI plus deterministic controls

Traditional automation is rules-to-action; generic AI is prompt-to-response. This platform's AI understands unstructured information, the policy engine applies organisation-defined rules, application code executes only permitted actions, and the audit layer records what happened — a more controlled architecture for enterprise use than either traditional automation or an unconstrained AI assistant alone.

26.10 Integration with the client's own ERP

Standard productivity suites are not built around any particular client's proprietary ERP. A connector layer allows approved ERP events (e.g. RFP → ERP customer lookup → account identification → assignment → task → response tracking → reporting) to join the same operational model — dependent on available APIs, documentation, authentication and client-side access.

26.11 Vendor-neutral architecture

The client keeps using Microsoft 365 (Outlook, Teams, SharePoint), WhatsApp Business, its ERP, CRM, browser and calendar as-is; this platform connects them into a unified management layer rather than requiring the client to replace its existing stack.

26.12 A local/private processing option — evaluated, not guaranteed

Cloud AI (the existing OpenAI setup) is used for the build and for rapid validation. For production, private/local LLM deployment can be evaluated against security requirements, hardware, model performance, latency, cost and data sensitivity — this is not a promise of complete local processing at this stage.

26.13 Comparison Table

Ratings below reflect each product category's general published capability as of August 2026, not an exhaustive audit of every plan/tier. "Varies" or "Available with configuration" is used wherever capability depends on licence tier, admin configuration, or the specific competing product.

Capability	Google Workspace + Gemini	Microsoft 365 + Copilot/Viva	Monitoring Platforms (e.g. ActivTrak)	Proposed Solution
AI email assistance	Strong	Strong	Not primary focus	Yes, policy-integrated
Email automation (routing/SLA)	Available with add-ons	Available with agents	Not primary focus	Core, policy-driven
Cross-platform integration	Primarily Google	Primarily Microsoft	Varies by vendor	Designed for multi-system
Teams meeting/activity data	Not applicable	Native (Viva Insights)	Available in some products	Planned connector
Browser/app usage tracking	Limited	Limited/varies	Strong — core capability	Yes
Employee productivity analytics	Limited/varies	Available via Viva	Strong — core capability	Yes, configurable per department
Email + activity analytics in one view	Not primary focus	Partial (Viva)	Not primary focus	Core capability
Client-specific policy engine	Workflows/ customisation	Workflows/Copilot agents	Configurable analytics/alerts	Core capability
RFP-specific workflow tracking	Custom build required	Custom build required	Not typically supported	Core workflow
Custom ERP integration	Via custom integration	Via custom integration	Rarely supported	Connector architecture
Natural-language management actions	Yes (Gemini/Workspace)	Yes (Copilot)	Increasingly available	Controlled, tool-gated agent
Screen recording / keystroke logging	Not applicable	Not applicable	Varies	Excluded by design
Local/private AI option	Product-dependent	Product-dependent	Product-dependent	Evaluated for production
Organisation-specific specialised agent	Via custom agent building	Via custom Copilot agents	AI insights in some products	Core product concept

26.14 Positioning Statement

The proposed platform is not intended to replace Google Workspace, Microsoft 365, Teams or existing employee productivity tools. Its value lies in acting as a specialised AI-driven management and orchestration

layer across the organisation's existing systems. It combines email intelligence, configurable business policies, employee activity analytics, productivity reporting and controlled natural-language actions into one organisation-specific operational agent.

The competitive advantage comes from the depth of client-specific workflow configuration, cross-system integration, policy-driven automation, a unified data model, and the ability to adapt the agent to the client's own business processes — not simply from the presence of AI features.

27. Business Benefits & KPIs

We do not commit to guaranteed percentage improvements without baseline data. The mechanism of each benefit is described below; actual figures should be measured against the client's current numbers once available.

KPI	How the Platform Affects It
Average email response time	AI triage, summarisation and policy-gated auto-reply reduce manual reading and drafting time.
SLA compliance	Deterministic SLA calculation and routing reduce missed/late responses.
Email backlog	Central dashboard surfaces pending items instead of leaving them buried in an inbox.
Automation rate	Tracked directly — % of emails resolved without manual drafting.
RFP response time	Explicit RFP detection and tracking closes a currently unmeasured gap.
Reporting effort	Monthly reports generated automatically instead of compiled by hand.
Productivity visibility	Cross-source usage data replaces anecdotal assessment with measured data.

Recommended baseline data to collect before go-live: current emails/day, current average response time, current hours spent on manual reporting. This lets both sides measure the improvement rather than assume it.

28. Full Project Scope

This is delivered as one unified engagement rather than separate phases. The Email Agent, Employee Management, RBAC, policy assignment and deterministic email/RFP KPIs make up the first build milestone. The Employee Activity Agent, Productivity Analytics, Reports, Voice Interface and Natural-Language Management Agent are part of the same committed scope, with each source (Teams, WhatsApp Business, Calendar, browser, desktop, ERP) going live as soon as its own client-side dependency — tenant consent, API approval, device rollout list, or credentials — is resolved, rather than waiting on a separate phase boundary. Items tagged Optional/Future (AI-generated trend narratives, cross-department configurable productivity definitions, local/on-prem LLM evaluation, best-effort active/idle time capture) are technically feasible but not committed or priced until requested.

29. Deliverables

Deliverable	Definition of Done
Extended Email Agent + Employee Management + RBAC + Policy assignment + Email/RFP KPIs	Built, demoed, automated tests passing, client sign-off.
Employee Activity Agent (each connected source), Productivity Analytics, Reports, Voice, NL Management Agent	Each source live once its client dependency is resolved; deployed and demoed as it lands.
Optional/Future items (trend narratives, cross-department productivity definitions, local/on-prem LLM evaluation, active/idle capture)	Scoped and priced only on request.

30. Development Effort

Effort is presented per module with its dependency and the assumption behind the estimate, so uncertainty is visible rather than hidden inside one total. Third-party-dependent items (Teams, WhatsApp, ERP) carry the widest ranges because access has not yet been confirmed.

Module	Deliverable	Effort	Dependency / Assumption
Requirements & architecture	Confirmed data model for expanded scope	2 days	None — can start immediately.
Employee Management + RBAC	Add/remove employees, role permissions, policy assignment UI	5 days	None.
Email Agent extension	RFP flag, richer policy assignment on top of the existing agent	3 days	Built as one agent from the ground up.
Email/RFP deterministic KPIs + dashboard	Overview + Email Command Centre + Reports (email-only) screens	5 days	None.
Integration Hub + normalised event model (core)	Adapter framework, event schema, health monitoring	4 days	None — built once, reused per source.
Teams adapter	Meeting/call metadata capture	3-5 days	Graph API/tenant admin consent — range reflects that uncertainty.
WhatsApp Business adapter	Business activity/messaging metrics	3–5 days	WhatsApp Business API approval.
Calendar adapter	Event/meeting metadata	2 days	Calendar API access.
Browser usage agent	Lightweight extension, domain/time capture	3-6 days	Confirmed browser list and rollout mechanism.
Desktop/OS usage agent	Lightweight collector, app/window/time capture	6-7 days	Confirmed OS list and device management approach.
ERP adapter	Client-defined events (e.g. RFP/order-related)	3-4 days	Entirely dependent on ERP API documentation quality —

			wide range until reviewed.
Productivity Analytics engine	Deterministic cross-source KPIs, configurable categories	5 days	Requires at least one activity source live.
Reports (full, cross-source)	Monthly report generation/export across all connected sources	2 days	Requires the Analytics engine.
Voice interface (TTS/STT)	Browser-based read-aloud and voice instruction capture	3-4 days	Chosen speech service and language/accent validation.
NL Management Agent (read-only)	Intent detection, tool-gated queries, audit	4 days	Requires the Analytics engine live.
NL Management Agent (actions)	Confirmation-gated actions via existing tools	2 days	Requires the read-only agent validated first.
Testing, security review & UAT	Test suite, isolation checks, sign-off, across every module	4 days	Ongoing across the build, not a one-off.

Total effort: approximately 59-70 on-days of work across the full unified scope, with the wide end of the range driven almost entirely by third-party API access timelines rather than build complexity — this should be re-estimated once ERP documentation and API approvals are in hand.

31. Dependencies

- Microsoft tenant admin consent and Graph API permissions for Teams and Calendar.
- WhatsApp Business API account and approval.
- ERP API documentation and credentials/sandbox access.
- Confirmed list of browsers and operating systems in active use, and an approved rollout mechanism for the lightweight extension/agent.
- Client infrastructure or hosting decisions if local/on-prem AI is pursued in production.
- Client legal/compliance sign-off on employee monitoring scope and jurisdiction-specific requirements.
- Client-defined data retention requirements for activity events and reports.
- Client input on department-specific productivity definitions, if the optional configurability is taken up.

32. Assumptions

- The Email Agent (Microsoft 365 OAuth, JWT auth, PostgreSQL, FastAPI, React, OpenAI, APScheduler) is built first and forms the technical foundation that every later module extends.
- “Activity tracking” means usage/time-spent metrics only, never continuous screen recording or keystroke logging, at any stage.
- Cloud-hosted AI (current OpenAI setup) is used throughout the build; local/on-prem deployment is evaluated separately for production and is not assumed by default.
- Pricing figures are placeholders pending the client's standard commercial terms and, for the ranged line items, pending confirmation of third-party access.

- Employees are informed that usage/time monitoring is in place for sources their employer has enabled, consistent with standard workplace-monitoring practice — exact disclosure mechanics are the client's responsibility.

33. Out of Scope

- Continuous screen recording, at any stage.
- Keystroke logging, at any stage.
- Monitoring of personal (non-Business-API) WhatsApp accounts.
- Full content monitoring of Teams messages or call audio/transcripts.
- Custom ERP-side development (as opposed to consuming existing ERP APIs).
- Hardware procurement, end-user device management, and on-device agent support beyond initial rollout.
- Legal/compliance review of monitoring practices in the client's jurisdiction — this is the client's responsibility, recommended before that capability goes live.
- Guaranteed fully local AI processing — evaluated for production only.

34. Risks & Mitigation

Risk	Impact	Mitigation
Third-party API access (Teams/WhatsApp/ERP) delayed or denied	Timeline and pricing ranges widen or specific adapters slip.	Dependency-gated deliverables; ranges converted to fixed prices once access confirmed.
Employee monitoring raises legal/compliance concerns	Activity capture cannot go live as planned.	Client legal sign-off required before that capability launches; usage/time-only design limits exposure.
Browser/OS fragmentation across client devices	Lightweight agents don't cover every device.	Confirm supported browser/OS list before build; scope agents to the confirmed list only.
ERP API documentation is incomplete or inconsistent	Effort estimate for the ERP adapter is unreliable.	Wide effort range flagged now; re-estimate once documentation is reviewed.
AI misclassification on the Email Agent	Incorrect routing or an unwanted auto-reply.	Safety gate and confidence threshold built in from day one; low-confidence items always route to human review.
Scope creep across ten modules	Timeline/budget pressure.	The classification matrix keeps each commitment explicit.

35. Implementation Roadmap

Delivered as one continuous build rather than separate phases — each item begins as soon as its own dependency (if any) is resolved, so the whole engagement does not wait on the slowest approval. Total delivery timeline: 12–14 weeks, depending on how quickly third-party access (Teams/WhatsApp/ERP) is confirmed.

Milestone	Indicative Timing
Extended Email Agent, Employee Management/RBAC, Email/RFP KPIs deployed and signed off	Weeks 1–4
Client secures Teams/WhatsApp/ERP access and confirms browser/OS list and legal sign-off (in parallel with the above)	Weeks 1–4
Integration Hub, Teams and Calendar adapters live, subject to access being resolved	Weeks 4–7
WhatsApp, Browser, Desktop and ERP adapters live as each dependency clears	Weeks 6–10
Productivity Analytics, Reports, Voice, NL Management Agent live	Weeks 9–14

36. Build & Validation Approach

Nothing in this proposal has been built yet — this section sets out how the Email Agent, as the first module, will be built and validated, so progress can be checked against a concrete plan rather than a vague promise.

The Email Agent will be built to cover the following capabilities:

- Microsoft 365 OAuth connection, supporting multiple users and multiple Microsoft 365 accounts at once, via Azure AD / Microsoft Entra ID.
- JWT login and encrypted token storage.
- Email fetching, parsing, and duplicate detection via Microsoft Graph.
- Contact management with known/unknown detection and classification.
- Configurable categories, teams, and policies.
- Thread-aware AI classification with confidence scoring.
- Safety gates for legal threats and fraud allegations.
- Policy-driven routing, SLA calculation, and task creation (with thread de-duplication).
- Controlled, policy-gated automatic replies.
- Full audit logging and a live dashboard.
- Complete multi-user data isolation.

Validation approach: development will follow an automated-test-first approach, with a full suite of automated tests run against a mock Microsoft 365 service before any code is connected to a real account. Once that automated suite passes, the same flows will be verified manually against a real Microsoft 365 tenant — real OAuth login via Azure AD, real inbox polling via Microsoft Graph, real token refresh, real incoming emails, real replies being sent, and several Microsoft 365 mailboxes running at the same time — before the module is signed off as complete.

This same build-then-validate pattern (automated tests first, then verification against the real client system) is applied to every later module as it is built, not only the Email Agent.

37. Conclusion

This proposal is built around one architecture — AI understands, policy decides, code executes, audit records — applied consistently module by module, from the Email Agent through to the full employee operations and

productivity agent, delivered as one unified engagement, built from scratch. Every requested capability is accounted for and classified as included, dependent on the client, optional, or explicitly excluded, so the client has a clear, honest picture of the complete platform and what it requires from their side to go live in full.

Appendix: Glossary of Terms

Plain-English explanations of the technical terms used in this document.

Term	In Plain Words
AI / LLM (Large Language Model)	The software (OpenAI) that reads an email or activity signal and understands what it's about — similar to how a person would read and summarise it.
API	A defined, secure way for two computer systems (e.g. this platform and Microsoft 365) to exchange data with each other.
OAuth	The “Sign in with Microsoft” style login (via Azure AD / Microsoft Entra ID). It lets the platform access an email account without ever seeing or storing the password.
JWT (JSON Web Token)	A secure digital pass issued when a user logs in, proving who they are on every subsequent action, and keeping each user's data separate from everyone else's.
Policy Engine	The rulebook the client configures — plain “if this, then that” business rules that decide routing, deadlines and whether an auto-reply is allowed. This is not the AI; it is ordinary, predictable code.
SLA (Service Level Agreement)	The deadline attached to a task — e.g. a VIP customer's email must be responded to within 2 hours.
RBAC (Role-Based Access Control)	Controls over who can see or do what in the platform — e.g. a Manager can see their own team's data but not another team's.
Audit Log	A permanent, timestamped record of every decision the system made, so anything can be reviewed later.
RFP	Request for Proposal — an inbound business inquiry the platform can specifically detect and track through to response.
Integration / Adapter	The piece of the platform that connects to one specific outside system (Microsoft 365, Teams, WhatsApp, the client's ERP, etc.) and translates its data into a common format.
Activity / Usage Data	How long someone spent on an app, site, or in a meeting — timing and category only, never the content, never a screen recording, never keystrokes.
KPI (Key Performance Indicator)	A measurable number used to track performance, e.g. average email response time or SLA compliance percentage.
ERP	The client's own core business system (e.g. for orders, customers, inventory) that this platform can optionally connect to, once access is granted.
Client Dependency	Something only the client can provide — an approval, an account, or access — before that specific piece of work can start.